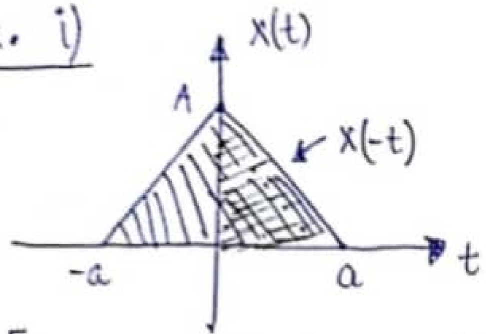
• Even and odd Signals.

Even signals

↳ Remain identical under folding operation. (Time Reversal)

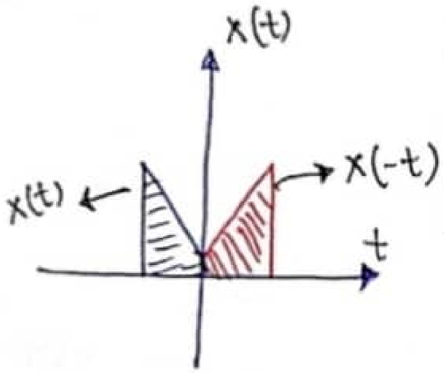
$$\boxed{x(t) \xrightarrow{T.R} x(-t) = x(t) \quad \forall t}$$

ex. i)



↳ On folding the above signal across y-axis we will get the same signal

(ii)



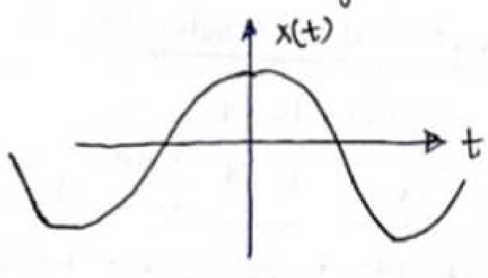
(iii) $x(t) = \cos\omega t$

$x(-t) = \cos(-\omega t)$

$= \cos\omega t$

$\therefore x(-t) = x(t)$

↳ Hence Even Signal.

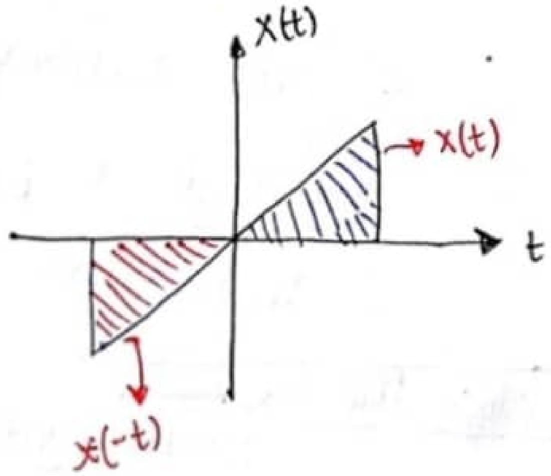


Odd signals

↳ Doesn't remain identical under folding operation.

↳ $x(-t) \neq x(t)$

↳ Antisymmetrical waveform about the y-axis.



$$\boxed{x(t) = -x(-t)}$$

Remember:

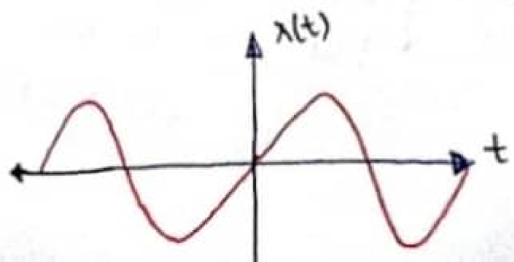
(i) At $t=0$; $x(0) = 0$
So odd signal is always zero at $t=0$.

(ii) The avg. value of any signal is always zero.

Example: $x(t) = \sin\omega t$

$x(-t) = \sin(-\omega t) = -\sin\omega t$

$= -x(t)$



- Even and odd components of a signal.

↳ Any CTS can be represented as sum of odd and even components.

↳ A general signal is neither even nor odd, but has both even and odd components.

$x(t) \longrightarrow$ Continuous time signal

$x_e(t) \longrightarrow$ Even component

$x_o(t) \longrightarrow$ odd component

$$\therefore \quad x(t) = x_e(t) + x_o(t) \longrightarrow (i)$$

Now put $t = -t$ [folding]

$$x(-t) = x_e(t) - x_o(t) \longrightarrow (ii)$$

Using (i) & (ii)

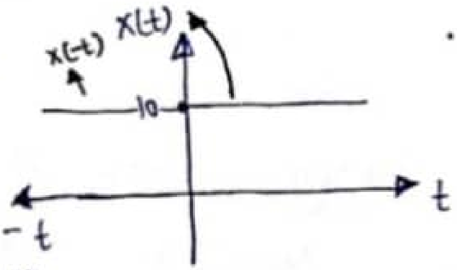
$$\boxed{x_e(t) = \frac{x(t) + x(-t)}{2}}$$

$$\boxed{x_o(t) = \frac{x(t) - x(-t)}{2}}$$

- Properties of Even and odd Signals

1. $x(t) = 10$

$x(-t) = 10$



$\therefore x(t) = x(-t)$

$\therefore$ $\boxed{\text{DC value is even signal}}$

↓ Purely even i.e $x_e(t) = 10$ and $x_o(t) = 0$.

2. Outcome of DC value + even signal

T.R → Time Reversal.

Ex: $x(t) = 10 + t^2$

(T.R) $x(-t) = 10 + (-t)^2 = 10 + t^2 = x(t)$

$\boxed{\text{DC value (even) + even signal = EVEN}}$

3. Odd signal + DC value

Ex: $x(t) = 10 + t^3$

T.R $x(-t) = 10 - t^3$ [neither even nor odd]

$\boxed{\text{DC value (even) + odd signal = Neither even nor odd}}$

4. Multiplication of two Even signals [E×E]

Example : $x(t) = t^2 \times t^4 = t^6$

$x(-t) = (-t)^6 = t^6 = x(t)$

$$\therefore \boxed{E \times E = E}$$

(Even signal)

5. Multiplication of two odd signals [O × O = E]

odd signal

Even signal.

Example :

$x(t) = t^3 \times t^5 = t^8$

$x(-t) = (-t)^8 = t^8 = x(t) \longrightarrow$ Hence Even signal.

6. Multiplication of O×E

Example : $x(t) = t^3 \times t^6 = t^9$

$x(-t) = (-t)^9 = -t^9 = -x(t)$

$$\boxed{O \times E = O}$$

7. $\frac{d}{dt}(E) = O$ odd signal (letter 'O')

Even signal

* The above property is not valid for DC values.

$\frac{d}{dt}(10) = 0$

(zero) → not odd signal.

8. $\frac{d}{dt}(O) = E$

odd signal.

Example. $\frac{d}{dt}(t^3) = 3t^2$

O E

9. $\int E \, dt = O$

Example : $\int t^4 dt = \frac{t^5}{5} + c$

(E) (O)

10. $\int O \cdot dt = E$

(odd signal)

11. $\frac{1}{O} = O$

(odd signal)

12. $\frac{1}{E} = E$

(Even signal)

Ques. Find the even and odd components of following signal :

$x(t) = \cos t + \sin t + \cos t . \sin t$

Sol. Method - I – (Using folding technique)

$\hookrightarrow (t \rightarrow -t)$

$x(-t) = \cos t - \sin t - \cos t \sin t$

$\therefore x_e(t) = \frac{x(t) + x(-t)}{2}$

$\boxed{\therefore x_e(t) = \cos t}$

$x_o(t) = \frac{x(t) - x(-t)}{2}$

$\boxed{x_o(t) = \sin t + \cos t \sin t}$

Method - 2 – (Using properties) $\longrightarrow$ (Shortcut method)

We Know, $\cos t \longrightarrow E$

$\sin t \longrightarrow O$

and $O \times E \longrightarrow O$ i.e $\cos t \sin t \longrightarrow O$

$\therefore x_e(t) = \cos t$, $x_o(t) = \sin t + \cos t . \sin t$.

Ques. Find even and odd components of the following signal.

$$x(t) = \underbrace{t^2 \sin t}_{1} - \underbrace{\frac{t^3}{\sin^2 t}}_{2} + \underbrace{t^3 \cos t}_{3} - \underbrace{\frac{\cos^3 t}{t^2}}_{4} + \underbrace{\frac{t^5}{\sin^5 t}}_{5}$$

Ans. **Method - 1** – (Using time reversal technique)

↳ $t \rightarrow -t$

↳ we find $x(-t)$

↳ $x_e(t) = \dfrac{x(t) + x(-t)}{2}$

↳ $x_o(t) = \dfrac{x(t) - x(-t)}{2}$

Method - 2 – (Using properties)

↳ $E \times E = E$

↳ $O \times O = E$

↳ $O \times E = O$

↳ $1/E = E$

↳ $1/O = O$

↳ $t^2 \rightarrow E$

↳ $t^3 \rightarrow O$

↳ $\cos t \rightarrow E$

↳ $\sin t \rightarrow O$

$$x(t) = \overbrace{\underbrace{t^2 \sin t}_{E \times O}}^{\text{odd}} - \overbrace{\frac{t^3 \rightarrow O}{\underbrace{\sin^2 t}_{E\,(O \times O)}}}^{\text{odd}} + \overbrace{\underbrace{t^3 \cos t}_{O \times E}}^{\text{odd}} - \overbrace{\frac{\cos^3 t}{t^2}}^{\text{Even}} + \overbrace{\frac{t^5 \rightarrow \text{odd}}{\sin^5 t}}^{\text{Even}}$$

$(E)^3 = E \quad 1/E = E$

$\sin^5 t$ ↳ $(O \times O \times O \times O \times O)$ → odd

$\therefore\; x_e(t) = \dfrac{-\cos^3 t}{t^2} + \dfrac{t^5}{\sin^5 t}$

$x_o(t) = t^2 \sin t - \dfrac{t^3}{\sin^2 t} + t^3 \cos t$